INSPECT-AI // QUALITY ASSURANCE REPORT

Universal AI Optical Metrology & Surface Inspection System | Generated: 2026-10-11 07:24:46 UTC

INSPECTION ID: f52b2dab-7fe8-4d80-8084-d5b6ee9e8f68	**OVERALL STATUS: FAIL**

Target Product:	Polymer Drive Bushing Ring	**Inspection Mode:**	COMBINED
Timestamp:	2026-10-11 07:24:46	**Unit of Measure:**	mm

Executive Summary & Decision Rationale:

Critical inspection criteria failed in: Dimensional Optical Metrology.

1. Dimensional Optical Metrology

Dimension	Expected	Measured	Deviation	Result
Length	52.00 mm	84.80 mm	+32.80 mm	**FAIL**
Width	52.00 mm	42.80 mm	-9.20 mm	**FAIL**

2. AI Visual Defect & Anomaly Analysis

Metric	Observed Value	Threshold / Criteria	Status
Anomaly Score	0.093	≤ 0.32	**PASS**
Reference Similarity	0.831	≥ 0.700	**PASS**
Detected Flaws Count	0	0 Permitted	PASS

3. Executed Quality Checkpoints

Check Name	Executed	Status	Details
Dimensional Optical Metrology	YES	**FAIL**	Dimensional out of tolerance: Length 84.80mm deviates by +32.80mm (permitted: -1.5 to +1.5); Width 42.80mm deviates by -
Deep Feature & Surface Defect Analysis	YES	**PASS**	No significant anomalies or surface defects detected. Anomaly score 0.093 is within permitted threshold 0.32.
Golden Reference Pattern Consistency	YES	**PASS**	Visual cosine similarity score with registered golden reference: 0.8305

Technical Disclaimers & System Boundaries:

- Planar 2D estimation assumes the object lies flat parallel to camera plane.
- Perspective distortion may affect boundary accuracy without multi-point camera calibration matrix.
- Shadows, high-glare lighting, or uneven background contrast can shift edge boundaries.
- Calibration relies on 5.00 pixels/mm scale factor.
- Unsupervised patch comparison relies on registered golden reference angles and lighting.
- Reflective specular highlights or shadow shifts can cause localized false-positive anomalies.
- Micro-defects smaller than 10 pixels may fall below feature map spatial resolution (8x8).

Inspection Algorithm: ResNet18 + OpenCV Metrology **Authorized Verification:** InspectAI System